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Solid Fuel

Experimental study on co-firing characteristics of ammonia with pulverized coal in a staged combustion drop tube furnace

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Abstract

Utilizing ammonia as a co-firing fuel to replace amounts of fossil fuel seems a feasible solution to reduce carbon emissions in existing pulverized coal-fired power plants. However, there are some problems needed to be considered when treating ammonia as a fuel, such as low flame stability, low combustion efficiency, and high NO_x emission. In this study, the co-firing characteristics of ammonia with pulverized coal are studied in a drop tube furnace with staged combustion strategy. Results showed that staged combustion would play a key role in reducing NO_x emissions by reducing the production of char-NO_x and fuel(NH_3)-NO_x simultaneously. Furthermore, the effects of different ammonia co-firing methods on the flue gas properties and unburned carbon contents were compared to achieve both efficient combustion and low NO_x emission. It was found that when ammonia was injected into 300 mm downstream under the condition of 20% co-firing, lower NO_x emission and unburnt carbon content than those of pure coal combustion can be achieved. This is probably caused by a combined effect of a high local equivalence ratio of NH_3/air and the prominent denitration effect of NH_3 in the vicinity of the NH_3 downstream injection location. In addition, NO_x emissions can be kept at approximately the same level as coal combustion when the co-firing ratio is below 30%. And the influence of reaction temperature on NO_x emissions is closely associated with the denitration efficiency of the NH_3. Almost no ammonia slip has been detected for any injection methods and co-firing ratio in the studied conditions. Thus, it can be confirmed that ammonia can be used as an alternative fuel to realize CO_2 reduction without extensive retrofitting works. And the NO_x emission can be reduced by producing a locally NH_3 flame zone with a high equivalence ratio as well as ensuring adequate residence time.

Keywords: Ammonia; Pulverized coal; NO_x emission, CO_2 reduction, Co-firing

1. Introduction

With global warming and climate changing becoming an urgent worldwide crisis, reducing CO_2 emissions and switching fossil fuel-based energy systems have become an international consensus. Many governments have declared to reduce CO_2 emission and determined to realize a carbon-zero or carbon-neutral society in the middle of this century. Accordingly, the low-carbon emission retrofit of the existing coal-fired power plant is indispens-

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able because coal is a type of cheap fossil fuel but generates more CO_2 than other fuels especially for developing countries. Various technologies have been proposed to reduce CO_2 emissions in thermal power generation process, such as oxy-fuel combustion [1], carbon capture and storage [2], and carbon-free fuel switching, etc. However, the feasibility and economic will be priorities for future retrofit and industry application.

Ammonia (NH_3) is preferred as a suitable hydrogen carrier because of its advantages of high energy density, easier transport, and well-developed infrastructures for storage and supply [3,4]. Therefore, utilizing ammonia as a co-firing carbon-free fuel in existing pulverized coal-fired power plants seems an attractive feasible solution to reduce CO_2 emissions. The co-firing characteristics of NH_3 with coal have been studied in national program of Japan, entitled “Gross-ministerial Strategic Innovation Promotion Program (SIP): Energy carriers” [5]. And the problems related to NH_3 combustion, mainly low combustion efficiency, combustion instability, and high NO_x emissions, have been discussed in the co-firing process of three-dimensional (3D) numerical simulations [6], a zero-dimensional (0D) simulation [7,8], and experiments using a 10 MW_{th} facility [9]. Results show that the NO_x emissions under the condition of 20% NH_3 co-firing could be decreased even lower than that of pure coal firing conditions through operating conditions adjustment, such as the velocity of NH_3 injection or staged air ratio. In addition, the NH_3 concentrations at the inlet of the stack were almost zero, indicating the supplied NH_3 is almost completely burned in the reactor. Thus, NH_3 co-firing with pulverized coal can be realized in the commercial boilers to reduce greenhouse gas emissions directly.

Furthermore, the injection methods of NH_3 are crucial for NO_x emissions. Tamura et al. [10] evaluated the characteristics of ammonia co-firing with three different ammonia injection methods using a coal-fired furnace with a capacity of 1.2 MW_{th}. And it was found that when ammonia was mixed in the pulverized coal nozzle with co-firing ratio below 30%, NO_x can be controlled at approximately the same level as the coal firing. While injection of ammonia from the side walls led to an increase of NO_x even for low co-firing ratios. The Central Research Institute of Electric Power Industry, CRIEPI, explored the effects of geometric changes of the ammonia side wall injection locations on the NO_x emissions through a 760 kW horizontal test furnace with a single burner [5]. Results showed that when ammonia was injected through the side port which is located at 1.0 m apart from the burner, NO_x can be lower than the level of coal combustion. While the NO_x concentration became higher as the injection position was shifted downstream, which was mainly because the reduced of residence time in the sub-stoichiometric zone for NO_x reduction.

However, it is noted that the results of previous works on the co-firing characteristics varied slightly on different burners. Furthermore, the deNO_x effect of NH_3 during the co-firing process is still not clear. In general, the correlative researches are inadequate since ammonia has not been considered as a main stream fuel until recently. Thus, more fundamental experiments are required to be conducted to find out the mechanism of interactions between different influencing factors. In this study, co-firing characteristics of pulverized coal and ammonia were studied in a staged combustion drop tube furnace. The effects of ammonia injection methods and downstream positions of staged air on the flue gas properties and unburned carbon contents are compared to find out a suitable NH_3 co-firing method allowing efficient combustion and low NO_x emission. In addition, the influences of co-firing ratio and reaction temperature are also considered to extend the basic knowledge of co-firing characteristics.

2. Methods

2.1. Experimental setup

The co-firing experimental system is shown in Fig. 1(a), which is consisted of a gas supply system, a loss-in-weight twin-screw micro-feeder coal feeder (K-TRON, America), an electrically heated furnace (Lenton, UK), an ash collected chamber, and an online FTIR gas analyzer (Gasmet FTIR DX4000, Finland). A vertical corundum tubular reactor was housed inside the furnace with an inner diameter of 75 mm and a length of 2000 mm. The specific description of the tubular reactor is illustrated in Fig. 1(b) through the section view. The pulverized coal was introduced directly into the hot reaction zone assisted with primary air through a jet nozzle (inner diameter of 10 mm) at the top of the reactor. Considering staged combustion, there are three staged air inlets (inner diameter of 6 mm) distributed around the center jet with different lengths. And the staged air can be fed downstream from the tip of the fuel jet at distances of 100, 200, and 300 mm, respectively. Each staged air tube is a 1 mm thick corundum pipe, and they are fixed on a 60 mm diameter circle centered on the center tube by a self-made steel plate. It is noted that the its bottom is sealed and the side outlet hole were against the furnace axis, benefiting for the mixing of staged air and flue gas. In the pre-experiments, it was found that the gas temperature profiles in length and radial directions of the middle reactor are totally stable and uniform, and the length of the isothermal reaction zone is around 450 mm. The staged air tube 1 is settled in the flame zone, the staged air tube 2 is settled in the post-flame zone, and the staged air tube 3 is settled in the high-temperature zone.

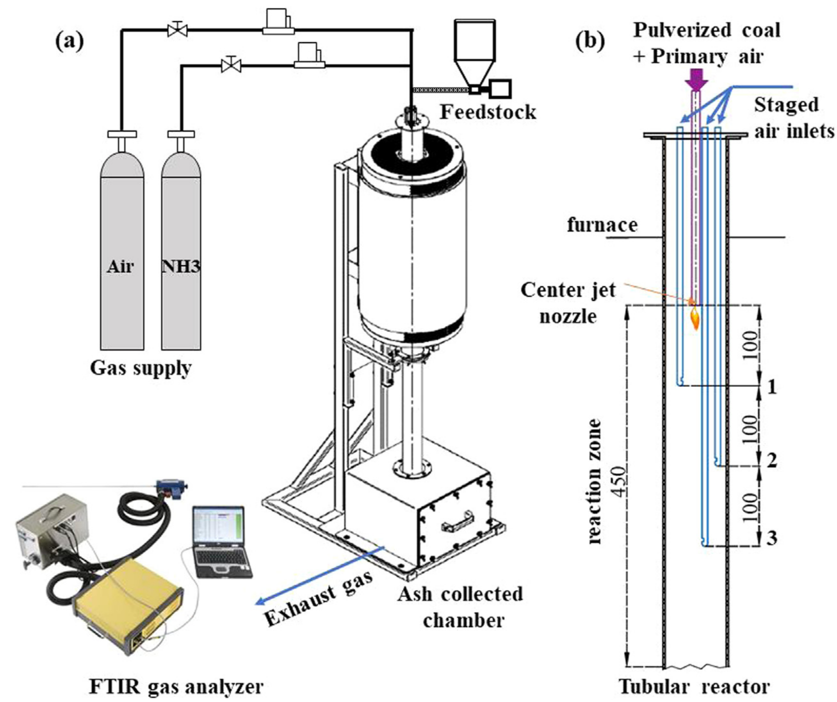

Fig. 1. Schematic diagrams of (a) the co-firing experimental system and (b) the sectional view of the tubular reactor.

Table 1
Proximate and ultimate analysis of the coal sample.

Raw coal	Proximate analysis (wt%)				$Q_{net,ad}$ (MJ/kg)	Ultimate analysis (wt%)				
	M_{ad}	A_d	V_d	FC_d		C_{daf}	H_{daf}	N_{daf}	$S_{t,daf}$	O_{daf}
Coal sample	8.77	12.11	31.31	56.58	25.02	80.67	4.50	1.07	0.41	13.35

Note: ad, on air-dried basis; d, on dry basis; daf, on dry ash free basis; St, total Sulphur; M, moisture content; V, volatile content; A, ash content; FC, fixed carbon; Q_{net}, low heating value.

2.2. Experimental conditions

The bituminous coal sourced from Inner Mongolia of China, was used in this study. The results of its proximate and ultimate analysis are shown in Table 1. Before using, the coals were air-dried and sieved to a particle size below 0.075 mm.

The feed rate of the coal is controlled and measured by the K-TRON micro-feeder, whose feeding error is within 5%. The feed gas was supplied from cylinder gas (air-99.999%, NH_3–99.99%), and the uncertainty of the gas flow rates controlled by several mass flow controllers (Alicat, 21–1–08–5 and 21S-1–32–1). Both considering the coal feeding rate of 20∼40 g/h and < 50% co-firing ratio of NH_3 by thermal input, the operational conditions are summarized in Table 2. The overall excess air ratio (the inverse of the equivalence ratio) is constant at 1.2. And the ratio of staged air is set to 30% of the total air flow when it comes to staged combustion. Accordingly, the total heat input is kept constant through fuel injection (278 W at 40 g/h coal case).

Table 2
Operational conditions of the co-firing experiments.

Operation parameters	Units	Values
Coal supply rate	g/h	20–40
Ammonia supply rate	g/h	0–27.03
Ammonia co-firing ratio	cal.%	0–50
Ammonia co-firing methods	–	Center Downstream
The distance shifted to downstream	mm	100, 200, 300
Rate of staged air	%	30
Total excess air ratio	–	1.2
Reaction temperature	°C	800–1200

When considering ammonia as a fuel for co-firing with pulverized coal, the injection methods of NH_3 are extremely important for itself combustion as well as NO_x reduction. In this study, three NH_3 injection methods were evaluated: 1) mixtures of pulverized coal, ammonia, and total air were injected through the center fuel nozzle, namely premixed mode; 2) the pulverized coal, ammonia, and primary air were injected from the center fuel nozzle, and amounts of stage air was supplied downstream of the reactor. Here use SC1∼SC3-NH_3Center/staged mode denoted staged air supplied cases, which means stage air supply distance of 100 mm, 200 mm and 300 mm downstream while NH_3 injection into the center nozzle; 3) the pulverized coal and primary air were injected through the center fuel nozzle, and the ammonia and staged air were injected into downstream, namely SC1∼SC3-NH_3Downstream/staged mode. In this study, the main experimental parameters are ammonia injection method, downstream position of staged air, co-firing ratio, and reaction temperature. Under the 1000 °C conditions, with a total gas flow rate of more than 5.09 L/min, the residence times are calculated in the range of 3.18 to 3.72 s in the reaction zone.

2.3. Data collection and analysis

For each experiment, flue gas properties such as O_2, CO_2, CO, NO, NO_2, N_2O, and NH_3 were continuously online analyzed at the furnace exit by an FTIR gas analyzer (Gasmet FTIR DX4000, Finland). Measuring ranges were set to be 0–25% for O_2 and CO_2, 0–2000 ppm for CO and NO, and 0–100 ppm for NO_2, N_2O, and NH_3. The uncertainty of the measurements were with 3% of the full measuring range of the analyzer.

It usually took 15–20 min to reach the equilibrium stage and the temporal stability of exhaust gas composition was monitored. For further evaluation, the average values were calculated from data recorded every five seconds for at least five minutes. Residual ash was collected and sampled from the bottle chamber, and their unburnt carbon contents (UC) were analyzed to indicate the burn out of pulverized coal. Noted that the residual ash was collected during the whole process including the initial unsteady prieod. The ash samples were burnt in an electrically muffle furnace at 815 ± 10 °C for more than 1 h. The weight loss through this procedure is assumed to be attributed to the combustion of unburnt carbon. Each experimental condition was repeated test for at least three times, and the average value was adopted to reduce system errors.

3. Results and discussion

3.1. Flue gas composition for different co-firing conditions

Fig. 2 shows the dynamic changes of flue gas compositions during the experimental procedure switching from pure coal combustion to 20% NH_3 co-firing at 1000 °C. Meanwhile, the influences of different ammonia injections: premixed mode, SC3-NH_3Center, and SC3-NH_3Downstream as

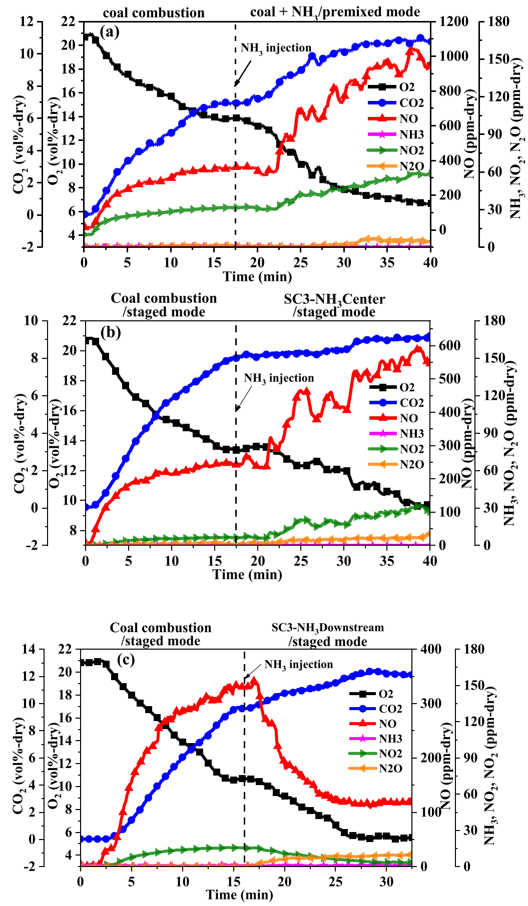

Fig. 2. Dynamic changes of flue gas composition during switching from pure coal combustion to 20% NH_3 co-firing at 1000 °C. (a) Coal combustion changes to Coal+NH_3/premixed mode; (b) SC3-Coal changes to SC3-NH_3Center/staged mode; (c) SC3-Coal changes to SC3-NH_3Downstream/staged mode.

mentioned above are compared. Noted that the supply rates of the air and coal during each continuous experiment were kept constant for the sake of operation and comparison. And, the excess air ratio for the coal combustion cases under this section would be over 1.2.

Fig. 2(a) shows that once the NH_3 was injected into the reactor after center premixed, the O_2 in the flue gas dropped to nearly 6.67% and CO_2 increased to about 10.78%. Meanwhile, the concentration of NO was increased dramatically to more than 1000 ppm in the flue gas. The main reactions related to NH_3 during co-firing conditions are expressed as below [11]:

$$2NH_3 + 3/2O_2 \rightarrow N_2 + 3H_2O \tag{1}$$

$$4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O \tag{2}$$

$$4NH_3 + 7O_2 \rightarrow 4NO_2 + 6H_2O \tag{3}$$

$$2NH_3 + 3NO \rightarrow 5/2N_2 + 3H_2O \tag{4}$$

$$4NH_3 + 3NO_2 \rightarrow 7/2N_2 + 6H_2O \tag{5}$$

Considering NH_3 is a carbon-free fuel, it can be anticipated that the substitution of coal by ammonia would directly lead to CO_2 reduction, while the increase of the CO_2 concentration shown in Fig. 1(a) is mainly due to the large formation of H_2O in the flue gas, which would be removed or condensed before entering into the FTIR analyzer. The drastic increase of NO and slightly increase of NO_2 are mainly produced by the fuel-NO_x originated from NH_3 compared with the case of pure coal combustion.

Staged combustion is a widely used technology to reduce NO_x by creating a reducing environment during early stage of coal combustion [12]. Therefore, it is considered to reduce the emission of NO_x by staged combustion in co-firing conditions. It is found in Fig. 2(b) that the concentration of NO was decreased to nearly 560 ppm, which is relatively less than that of premixed mode. It was noticed that the CO_2 concentration changed slightly after NH_3 injection, and more O_2 was left in this case due to the un-sufficient combustion. Therefore, less carbon was consumed under this situation, which also means the less production of char-NO. Moreover, the clearly decrease of NO under staged combustion also indicated that the formation of fuel(NH_3)-NO was also compressed under the fuel-rich condition. The effects of staged combustion ratio on co-firing were also evaluated by Ishii et al. [9], and they found that the distribution of NO concentration could be even lower than that of pure coal combustion by choosing a proper staged combustion ratio.

It is well known that NH_3 owns a reasonable prediction in decreasing the emission of NO_x by the reaction of selective non-catalytic reduction mechanism (SNCR) through R4-R5. When the NH_3 was injected downstream (shown in Fig. 2(c)), only the pulverized coals were burned under the air-rich conditions in the region from the center jet nozzle to the downstream outlet of staged air. So when the NH_3 was injected into the low-O_2 and high-NO reaction tube, the NH_3 would reduce NO to N_2 efficiently and the NO concentration began to decline to even lower than that of coal combustion.

It was also found that more amount of O_2 is consumed and the CO_2 rose slowly in this case. Thus, O_2 is consumed not only for char oxidation, but also for the reactions related to NH_3 dissociation and oxidation. What calls special attention is that the NO_2 concentration decreased while N_2O content increased when NH_3 was injected downstream. It was reported in the study of Elbaz et al. [13] that the formation of N_2 would be promoted through the N_2O and NNH intermediate channels, which is called thermal De-NO_x processes. Thus, it can be concluded that in the low-O_2 and high-NO downstream zone, the generation of fuel-NO was effectively suppressed.

As for NO_2, it was noticed that its change trends were identified with those of NO. And the NH_3 downstream injection would also contribute to the reduction of NO_2 emissions. Special attention should also be paid to the release of N_2O during co-firing conditions because it has a global warming potential of as high as 310 times than that of CO_2 [14]. The emissions of N_2O during pure coal combustion were pretty much zero. While the contents of N_2O produced from co-firing conditions were a little higher than those of pure coal combustion, which were probably ascribed to the abundant fuel-N content of NH_3. Moreover, the N_2O emission was increased to a maximum value in the case of SC3-NH_3Downstream/staged mode. However, the maximum value of N_2O emission was found all below 10 ppm in this study, which cannot be a major obstacle for ammonia co-firing.

From the point view of safe and stable operation consideration, NH_3 leakage should be avoided. In this study, when the NH_3 was injected into the center fuel jet nozzle, NH_3 emissions were almost zero. As for the NH_3 injection downstream case, NH_3 emissions increased slightly, but is still in a very low level, i.e. <1 ppm. Hence, it is indicated that the supplied NH_3 is almost completely burned in all the testing cases. The co-firing of NH_3 with pulverized coal is considered a feasible solution for reduction of CO_2 emission in existing coal fired facilities.

3.2. The effect of ammonia co-firing methods on co-firing performances

Fig. 3 shows the flue gas properties and unburned carbon content in residue ash under different NH_3 injection methods with 20%-thermal input of ammonia and temperature of 1000 °C. Furthermore, when adopting staged combustion strategy, the effects of the position of staged air entered into the reactor were also studied by changing the staged air inlets as mentioned above. It was shown that the substitution of coal by ammonia would directly lead to CO_2 reduction. And the reduction of CO_2 is nearly the same for different co-firing cases, which is approximately 20% in the 20% ammonia co-firing conditions.

In the case of NH_3 premixed mode (shown in Fig. 3(a)), the NO_x concentration increased to 1043.65 ppm, which is much higher than that of 277.44 ppm for the pure coal combustion due to much more fuel-NO_x derived from NH_3. The CO emission was decreased by 0.56 times in co-firing conditions compared to coal combustion. Except for the reduction of coal supply, the more abundance of H_2O in the reaction atmosphere due to the higher H and OH radicals coming from oxidization

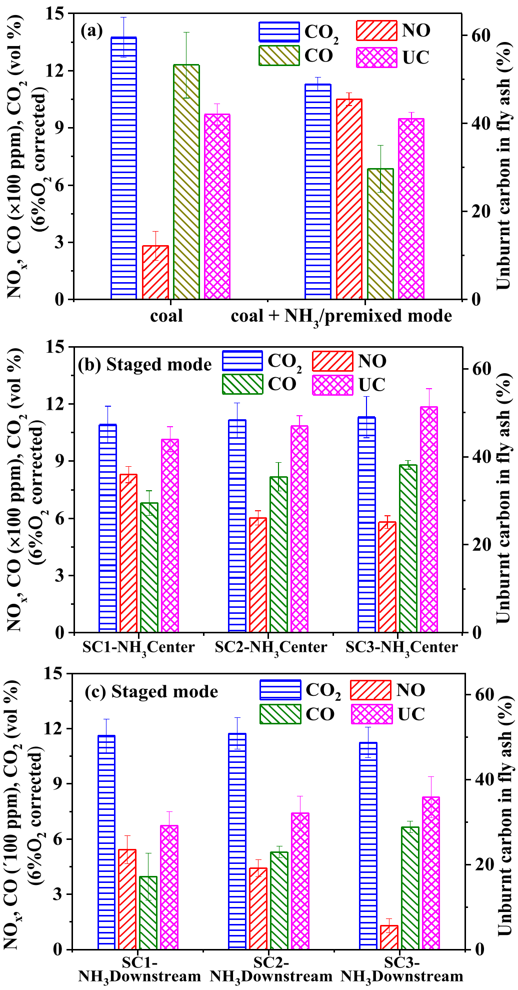

Fig. 3. The effects of different NH_3 co-firing methods on the co-firing emission performances (co-firing ratio:20%, temperature: 1000 °C).

of NH_3 may contribute to the lower CO emissions according to the reaction of R6.

$$CO + OH \rightarrow H + CO_2 \tag{6}$$

And the content of UC in residue ash obtained from premixed mode is slightly smaller than that of coal combustion. The numerical studies made by Ishihara et al. [7] have reported that NO concentration would reach the maximum in the flame zone (nearly 3 times higher than that of coal combustion) when NH_3 was injected into the burner with combustion air, afterwards it would decrease until over-fire zone due to reduction by char (R7).

$$C + NO \rightarrow 0.5N_2 + CO \tag{7}$$

As shown in Fig. 3(b), it was found that the NO_x concentration decreased when using staged combustion strategy. Furthermore, NO_x decreased continuously from 829.20 ppm to 579.68 ppm as the outlet of the staged air was shifted downstream. As mentioned above, the decrease of NO_x originated from the fuel(NH_3)-NO_x, which is affected by the local equivalence ratio. It was found when the NH_3 was injected into the center jet under staged combustion cases, the reaction zone from the tip of the fuel jet to the outlet of staged air is estimated to be under fuel-rich condition. In fuel-rich NH_3-air flame, NO_x concentration would decrease with the increase of equivalence ratio [5]. As the distance of the staged air outlet enlarged away from the top, there would be comparatively more NH_3 exist in the reaction zone, which also help enrich the local equivalence ratio. Hence, it can be explained the reason for decreasing NO emission with the increasing downstream distance of staged air. However, at that time, the combustion of pulverized coal would be aggravated, resulting in the increase of CO concentration and a higher level of unburnt carbon content.

Fig. 3(c) shows that in the case of NH_3 injection downstream with staged air, the concentration of NO_x became even lower than the cases of NH_3 injection into center under staged combustion conditions. Furthermore, the NO_x concentration would keep decreasing as the NH_3 injection position was shifted downstream, and the minimum NO_x concentration can reach to 128.63 ppm, which is lower than that of 277.44 ppm for pure coal combustion. These findings can also be explained by the NH_3/air equivalence. It was found that in the case of SC3-NH_3Downstream/staged mode, the local equivalence for NH_3/air would be the highest in the vicinity of the staged air outlet, which would suppress the production of fuel(NH_3)-NO_x. Together with the deNO_x effect of NH_3 downstream, the lower NO_x emission than the coal combustion can be achieved. Similar findings have also been reported by Tamura et al. [9] that the equivalence ratio in the burner zone appears to present a border line for NO_x emission, and it is possible to obtain strong NO_x reduction under NH_3 rich conditions. Meanwhile, it can be found that the unburnt carbon contents and CO concentration in the case of NH_3 injection downstream are lower than those of other firing conditions. This is probably because there are only pulverized coals were burnt under the air-rich conditions in the region from center jet nozzle to downstream outlet of staged air. And it can be concluded that when the NH_3 was injected downstream with staged air, NH_3 would play a role both as a reducing agent and a fuel for effective heat release.

3.3. The effect of ammonia co-firing ratio on co-firing performances

It can be anticipated that a higher NH_3 co-firing ratio will be considered to broaden the range of flammable fuels and achieve more reduc-

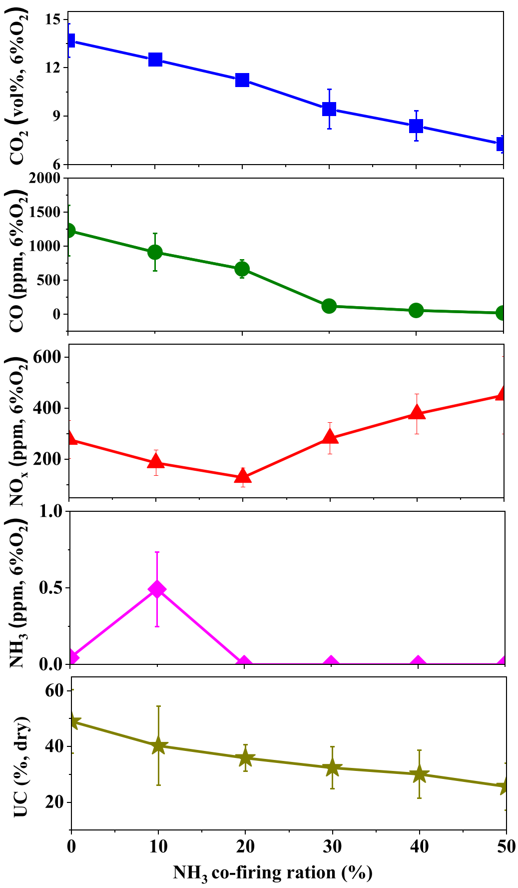

Fig. 4. Effect of the NH_3 co-firing ratio on flue gas components and UC contents. (injection method:SC3-NH_3Downstream/staged mode, temperature: 1000 °C).

tion of CO_2 emission in the future [4]. In this study, the effect of NH_3 co-firing ratio varying from 0% to 50% on the flue gas compositions and UC contents are studied in the case of SC3-NH_3Downstream/staged mode under 1000 °C, the results are illustrated in Fig. 4. Since ammonia is a carbon-free molecular, CO_2 emissions can be decreased in proportion to the increase of co-firing ratio as expected. As for NH_3 emissions, the maximum value was observed to be 0.5 ppm for the case of 10% ammonia co-firing, and other conditions were all keep almost zero. These emissions are sufficiently small and the supplied NH_3 could be considered almost completely burned in the reaction zone.

The concentration of CO also decreased with the increase of co-firing ratio, which is estimated to be caused by the reduction of coal supply and the much easier occurrence of gas phase oxidation (R6) in the NH_3 burned gasses. And when the co-firing ratio reached above 30%, the CO emission can be kept within a very low level of ∼100 ppm.

As for NO_x, it first decreased until the co-firing ratio reached to 20%, afterwards it increased monotonically with the increase of co-firing ratio. And when the co-firing ratio was below 30%, the NO_x emission can be controlled at approximately the same level as the pure coal combustion. And the NO_x increased to 451.07 ppm in the case of 50% co-firing. This trend of NO_x emissions with the NH_3 ratio is consistent with the results of the 0D calculation obtained using the ANSYS Chemkin-Pro [7]. It was found that when the co-firing ratio was higher than 40%, the reactions related to NH_3 decomposition tended to not involve NO. Such reactions are known as rich NH_3 chemistries, which are characteristics in rich and low-temperature NH_3 flame. Due to these reactions, fuel(NH_3)-NO_x production tends to decrease in the location where NH_3 reacts, thus resulting in the slow increase of NO_x with the co-firing ratio. However, it should be noted that as the NH_3 co-firing ratio increased, the location where NH_3 reactions occurred would migrate downstream. Therefore, special attention should be paid to selecting the downstream injection positions for NH_3, because if the residence time is too short for NO_x decomposition, the same effect would be discount.

The content of UC decreased with the increase in co-firing ratio. In the case of NH_3 injection downstream, the supply of the coal was reduced for higher co-firing ratio conditions. Therefore, the excess air ratio between the tip of the fuel jet and downstream staged air outlet would be increased with the increase of co-firing ratio, which increased to 1.68 for the case of 50% ammonia co-firing. This is considered to contribute to lower unburnt carbon levels.

3.4. *The effect of reacting temperature on NO_x concentrations*

NO_x emission is strongly affected by fuel-nitrogen (fuel-N) and combustion temperature [15]. Fig. 5 shows the effects of reaction temperature on the co-firing characteristics, where the ammonia co-firing rate is constant at 20% and the NH_3 injection method is SC3-NH_3Downstream/staged mode. It can be seen that the CO_2 emission under 20% NH_3 co-firing was independent of temperature, which is decreased by about 0.8 times compared with the case of coal firing. The CO concentration decreased monotonically with the increased temperature because the R5 was favored by temperature.

As for NO_x, it can be found that its emissions were strongly affected by the temperature. The NO_x emissions under 1000 °C were lower than the case of coal firing, which was 149.92, 32.93, and 128.63 ppm for 800, 900, and 1000 °C, respectively. However, when the temperature was higher than 1000 °C, the NO_x emissions increased significantly, which can go up to 1117.57 ppm at 1200 °C.

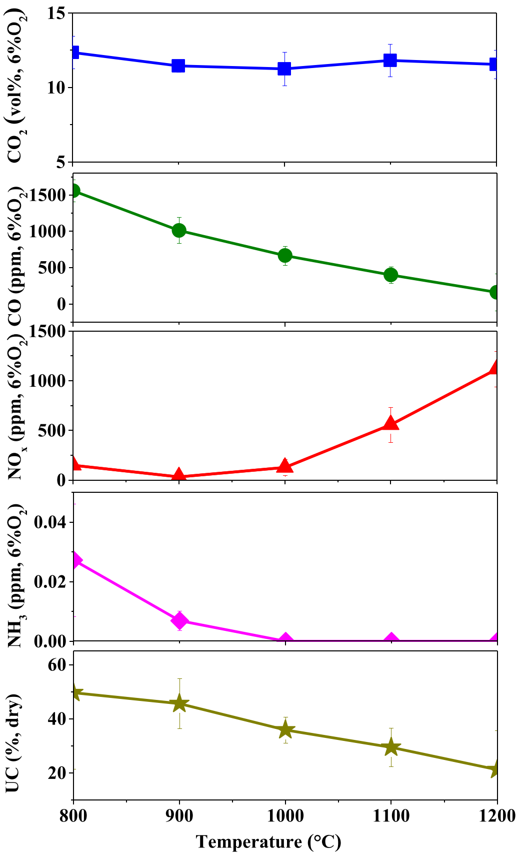

Fig. 5. Effect of the reacting temperature on normalized CO_2, CO, NO_x, and unburned carbon in ash (UC). (injection method:SC3-NH_3Downstream/staged mode, co-firing ratio:20%).

It was reported by Lyon et al. [16] that NO_x reduction in flames can be enhanced using NH_3 in the presence of O_2 within a specific temperature range of 900–1100 °C. The highest NH_3 emissions are calculated as 0.027 ppm at 800 °C, which is still sufficiently small as emission. Furthermore, since the unburnt carbon content decreased with the temperature, more consumption of char during co-firing processes might add to more production of char-NO_x.

It seems that the optimum reaction temperatures for the 20% NH_3 co-firing are in the range of 900 to 1000 °C. However, this range might be too narrow to be widely used. According to the above discussion, the NOx production derived from NH_3 (fuel(NH_3)-NO) can be inhibited by producing a locally NH_3 flame zone with a high equivalence ratio as well as ensuring an adequate residence time. Thus, the total excess air ratio as well as the proportion of the staged air can be adjusted to achieve a lower NO emission even at higher temperatures.

4. Conclusion

The experimental study on co-firing characteristics of ammonia with pulverized coal are performed in a staged combustion drop tube furnace. The influences concerned about the NH_3 injection method, the downstream outlet of staged air, co-firing ratio of NH_3, and reaction temperature on the flue gas composition and unburned carbon content are analyzed in detail.

Results show that the substitution of coal by NH_3 would directly contribute to the reduction of CO_2 emission. And the staged combustion would play a key role in decreasing NO_x emissions by reducing the production of char-NO_x and fuel(NH_3)-NO_x simultaneously. Furthermore, when the NH_3 was injected into 300 mm downstream under the condition of 20% co-firing, the highest local equivalence ratio in the vicinity of the location of NH_3 injection would suppress the production of fuel(NH_3)-NO_x. Meanwhile, the denitration effects of NH_3 behave more prominent in the low-O_2 and high-NO downstream zone. These integrating effects contribute to its lowest NO_x emission, which is even lower than that of coal combustion.

The increasing co-firing ratio would lead to the descending CO_2 emission and unburnt carbon content in the residue ash. NO_x emissions can be kept at approximately the same level as pure coal combustion when the co-firing ratio is below 30%. And the influences of reaction temperature on NO_x emissions are closely associated with the denitration efficiency of the NH_3. Almost no ammonia slip has been detected for any injection methods and co-firing ratio in the studied conditions. Thus, it can be confirmed that ammonia can be used as an alternative fuel to realize CO_2 reduction without extensive retrofitting work. And the NO_x can be reduced by producing a locally NH_3 flame zone with a high equivalence ratio as well as ensuring adequate residence time.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

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